

UCE858 : PAVMENT ANALYSIS AND DESIGN				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: This course equips students with a comprehensive understanding of pavement materials and structural design. It covers the properties and testing of aggregates, bituminous binders, and concrete, along with mix design methodologies. Students will learn pavement design principles for flexible and rigid pavements using IRC guidelines, traffic load analysis, and mechanistic-empirical methods. Hands-on experience with software tools will enhance analytical skills, enabling students to design, evaluate, and optimize pavement systems effectively.				
Detailed Syllabus- 1. Pavement Materials—Review of aggregates (properties, classification, tests, aggregate blending, alternative aggregates); review of bituminous binders (grades, types, modified binders, tests, rheology); review of soil-aggregate mixes; materials for rigid pavements; mix design for asphalt mixtures; mix design for pavement quality concrete as per IRC 44. Software for aggregate blending. Rheological studies for modified binders using DSR. 2. Flexible Pavement Design and Analysis—Concepts of VDF, LDF, CFD analysis; traffic calculations; provisions from relevant IRC codes; design as per latest IRC 37 guidelines, design of pavements using cement treated subbase, cement treated bases, Granular crack relief layer (AIL), Stress absorbing membrane inter layers, Full Depth Reclamation & bitumen emulsion/foamed bitumen treated reclaimed asphalt pavement RAP base & IRC approved new technologies; Low volume pavement design using IRC: SP-72 guidelines, Pavement design using geosynthetics as per IRC: SP: 59 guidelines, Design procedures in case of construction using recommended recycling technologies using IRC:120 guidelines. 3. Rigid Pavement Design and Analysis—Types of rigid pavements; Stresses in rigid pavement; relative stiffness; combination of stresses; IRC 58 and IRC SP 62 recommendations; joints; load transfer by dowel bars, tie bars using software/spreadsheet, Design of PQC slabs using separation membrane and without separation membranes, Short panelled concrete slabs design. Special Topics—IRC:15; IRC: SP:76; IRC:SP:63; perpetual pavements; PCI analysis, software exposure to KENPAVE, KENSLABS; PerROAD; industry case studies				
Exercise/ Assignments - Develop spreadsheet / computer program for asphalt mix design, mix design of pavement quality concrete, calculations of VDF, design traffic, etc. - Apply soft computing techniques on pavement data available in open repositories				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: - Evaluate pavement materials and mix designs using standard tests - Design and analyze flexible pavement systems using relevant guidelines - Design and analyze rigid pavement systems using relevant guidelines				
Text Books - Pavement Analysis and Design, Huang, YH, Prentice Hall, 2008 - Principles of Pavement Design, ET Yoder and MW Witczak, John Wiley and Sons, 1975. - Highway Engineering, SK Khanna, CEG Justo, A Veeraragavan, Nem Chand & Bros, 2015.				

4. Design and Performance of Road Pavements, D Croney and P Croney, McGraw Hill, 1998

Reference Books

1. IRC:37-2018, Guidelines for the design of flexible pavement
2. IRC:58-2015, Guidelines for the design of plain jointed rigid pavements for highways

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25